

The predictions for the artificial applicants are not entirely unexpected when considering the structure of the dataset. Although the first applicant has stronger academic metrics (higher GPA and GRE scores), they are applying to a PhD program and are classified as an international applicant. Both of these factors are likely associated with lower acceptance rates in the dataset.

In contrast, the second applicant, despite having lower academic metrics, is applying to a Master's program and is classified as a local applicant. These factors are typically associated with higher acceptance rates.

This suggests that the model is capturing broader patterns in the dataset, such as the relative difficulty of PhD admissions and differences between international and local applicants, rather than relying solely on academic performance metrics.

Since we had to replace numerous GRE and GPA values with median values during reprocessing, the influence of these numerical features may be reduced. As a result, categorical features like degree type and applicant origin may have a stronger impact on predictions. There may be some degree of multicollinearity.